Supplementary Information — NT-BE / X2-ω

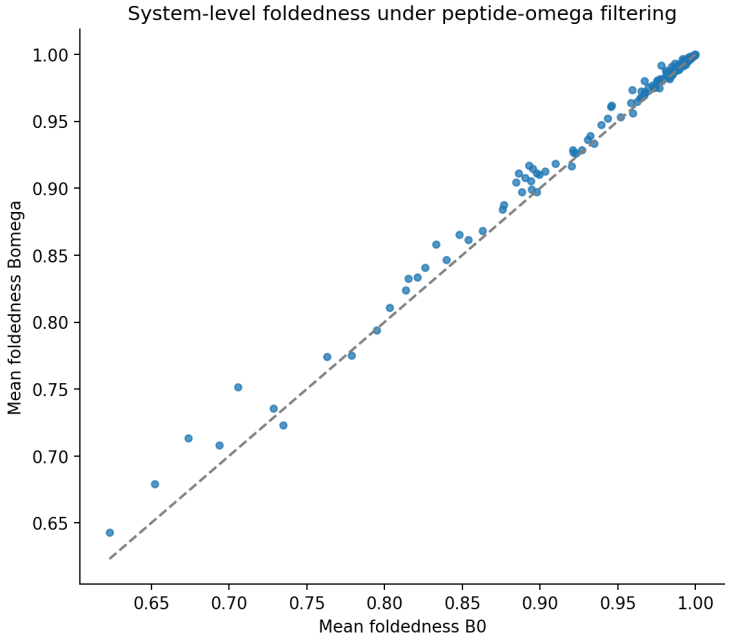
Fixed study data and provenance; complete machine-readable source tables are included in this package.

S1. Input provenance and environment

Fixed BioEmu stability archive: 154 systems (42 WT, 112 mutant), 112 paired mutant-minus-shared-WT estimates, official saved A ensemble. BioEmu benchmark source Zenodo 15672282; 153+1 recovered dataset lineage is preserved in the fixed Publication Gate. Original historical dependency versions are incompletely documented.

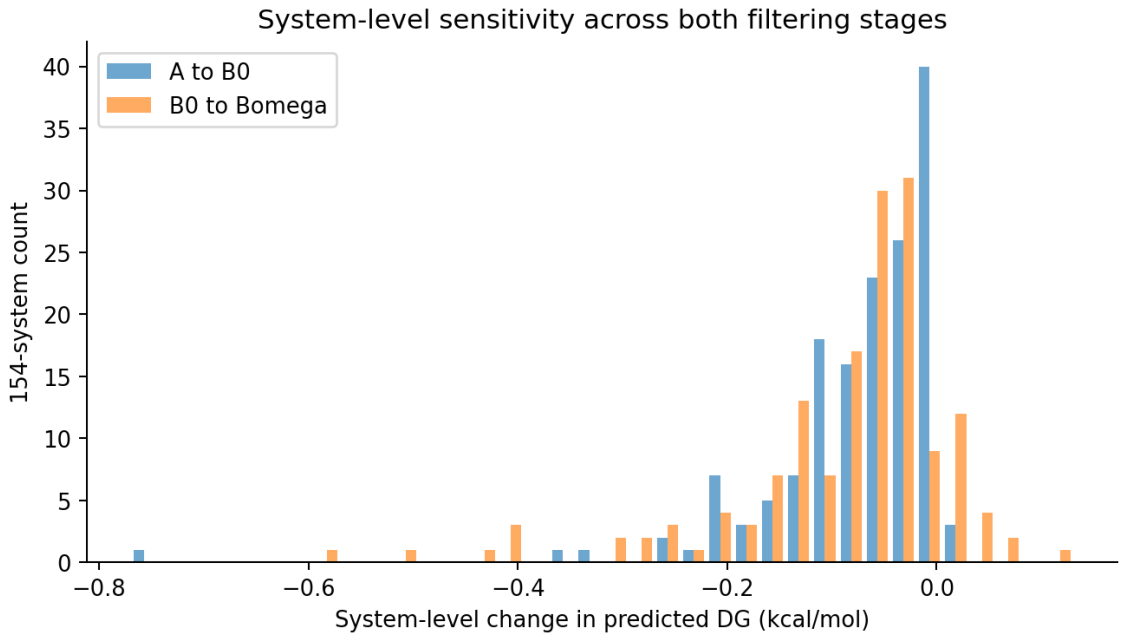
S2. Frame retention and system-level foldedness

A/B0/Bω saved-frame counts: 302,800 / 298,188 / 272,381. For 154 systems, Bω–B0 foldedness shift median +0.001850, range −0.011552 to +0.045725; 131 positive and 23 negative. Exact per-system values: supplement/Table_S2_S3_system_foldedness_dG.csv.



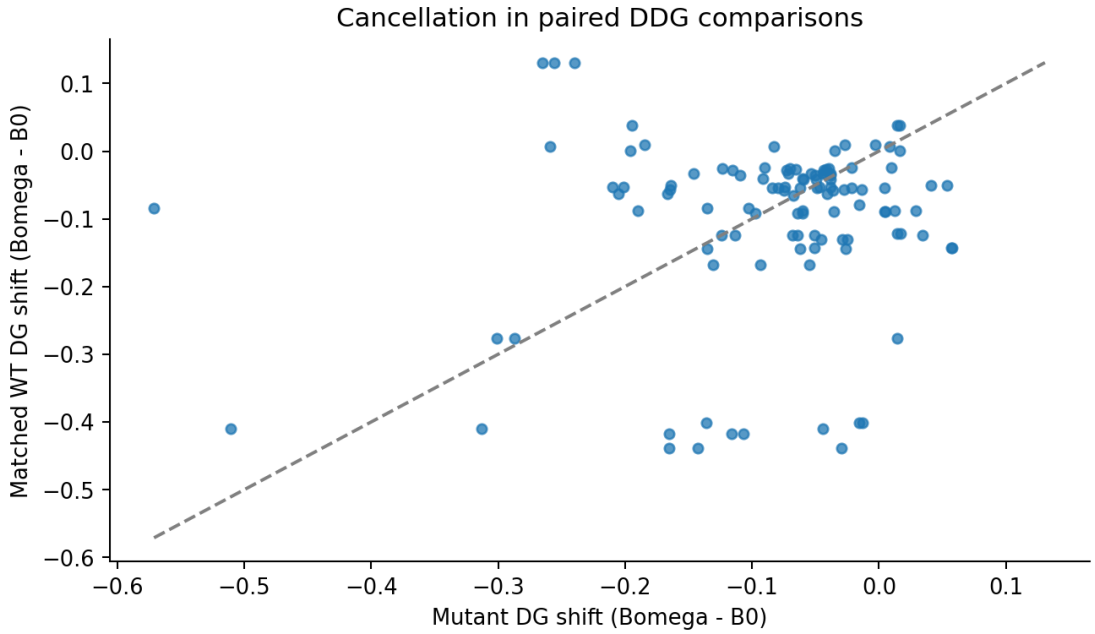
S3. System-level ΔG sensitivity

Complete A, B0 and Bω system-level ΔG and signed stage changes appear in the joined 154-system CSV. FNCSettings: contact_lambda=1.2 and contact_delta=0.0; 295 K transform. Neither these ΔG values nor their differences are experimental free energies.



S4. Pairwise ΔΔG decomposition

The 112-row Table S4 includes mutant and shared-WT contributions, three ΔΔG stages, stage differences, experimental references and threshold flags. Counts above the descriptive |0.1 kcal/mol| shift: A→B0 19/112, B0→Bω 40/112, overlap 12, union 47, and cumulative A→Bω 44.



S5. Boundary diagnostics

Parameter	Frozen observation
ω cutoff	Strictly >30° deviation from the appropriate cis/trans reference
Boundary	292 B0 frames within ±0.02°; tested alternatives yield 39/112 vs frozen 40/112
Unknown	0 frames
Under-70	Three systems defined versus target frames, not B0 frame count

S6. Targeted original-coordinate verification

Audit layer	Evidence
Saved NPZ / 154 systems	All masks, output arrays and arithmetic checked
Removed-frame QA	4,443 excluded frames across nine selected systems
Source-coordinate QA	10 selected systems; 2,700 frames; not all 154 source XTC datasets

S7. Verification boundaries and data index

The stored A archive is not the unsaved raw-generative model population. Shared-WT pairs are not independent replicates. The X2-ω gate did not establish general filter superiority or physical equilibrium recovery. Exact files and hashes: provenance/VALIDATION_REPORT.json and SHA256SUMS.txt.

Distributed file	Content
supplement/Table_S2_S3_system_foldedness_dG.csv	154 system foldedness, ΔG, retention and stage differences
supplement/Table_S4_cross_stage_112_pairs.csv	112 matched mutant/WT ΔΔG comparisons
supplement/Table_S5_boundary_and_omega.csv	Threshold and ω frame metadata
data/all_npz.zip	154 archived system output NPZ members